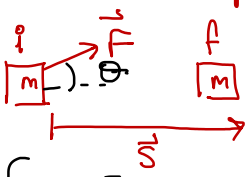
04 Work and Energy

Thursday, November 30, 2023 1:36 PM

Ch6 & Ch7

work, KE — PE, Energy cons.

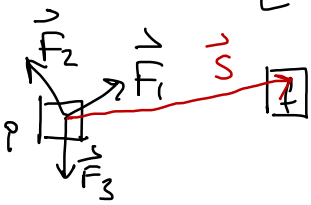
work = Force · displacement



$$W = \vec{F} \cdot \vec{s} = F s \cos\theta$$

$$W = F\,s = [N\ m] = [\text{Joule}]$$

$$\left[kg\,\frac{m}{s^2}\,m\right] = \left[kg\,\frac{m^2}{s^2}\right] = J$$



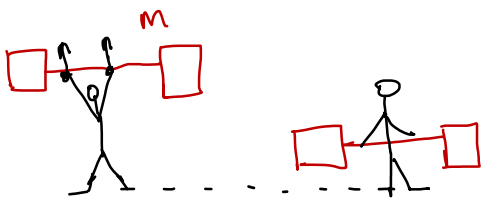
$$W_{F_3} = \vec{F}_3 \cdot \vec{s}$$
$$W_{F_2} = \vec{F}_2 \cdot \vec{s}$$
$$W_{F_1} = \vec{F}_1 \cdot \vec{s}$$

Total work

$$\Sigma W = W_{F_1} + W_{F_2} + W_{F_3} = \vec{F}_1 \cdot \vec{s} + \vec{F}_2 \cdot \vec{s} + \vec{F}_3 \cdot \vec{s}$$

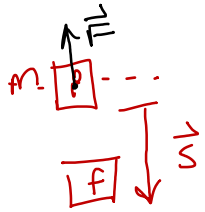
$$\Sigma W = \underbrace{\Sigma \vec{F}}_{\text{net force}} \cdot \vec{s}$$

ex) weightlifter & dumbell



* work done by the person onto the dumbell

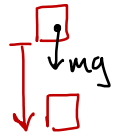
$$F = mg$$



$$W = \vec{F} \cdot \vec{s} = F s \cos 180 = -F s = -mgs$$

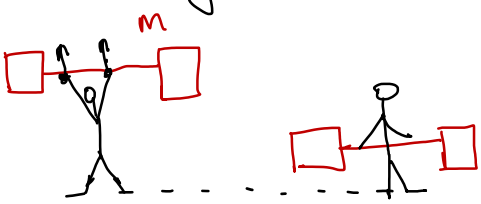
* work done by grav. force onto the dumbell

$$\Sigma W = mgs - mgs$$
$$\Sigma W = 0$$



$$W_{mg} = m\vec{g} \cdot \vec{s} = mgs$$

work done by dumbell onto the weightlifter.



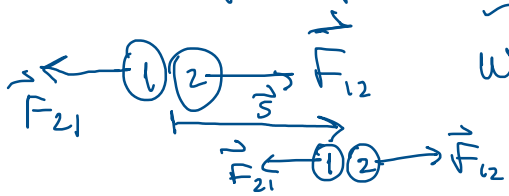
action reaction pairs

$$W_{DW} = \vec{F}_{DW} \cdot \vec{s} = -(\underbrace{-mgs}_{W_{WD}}) \qquad \vec{F}_{DW} = -\vec{F}_{WD}$$

DW: dumbell → weightlifter

$$W_{DW} = -W_{WD}$$

* action reaction pair forces DO OPPOSITE sign work!



$W_{12} = -W_{21}$

ex) $\vec{F} = (160\hat{i} - 40\hat{j})$ N $W = ?$ $W = \vec{F}\cdot\vec{s}$ $\hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{i} = 0$

$\vec{s} = (14\hat{i} + 11\hat{j})$ m $= 160(14)\,\hat{i}\cdot\hat{i} - 40(11)\,\hat{j}\cdot\hat{j}$

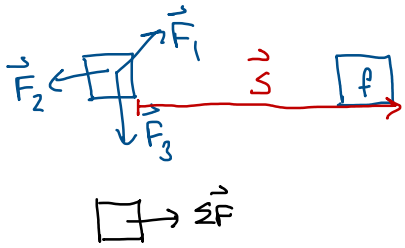
what's the angle between $\vec{F}$ & $\vec{s}$? $W = 1800$ J

$$\vec{F}\cdot\vec{s} = W = Fs\cos\overset{?}{\theta}$$

$$1800 = \underbrace{\sqrt{160^2 + 40^2}}_{210}\,\underbrace{\sqrt{14^2 + 11^2}}_{18}\cos\theta \Rightarrow \theta \cong 30^\circ$$

$$\theta = \cos^{-1}\left(\frac{1800}{210(18)}\right)$$

WORK and Kinetic Energy Theorem



$\Sigma W = \Sigma\vec{F}\cdot\vec{s}$

Ch4 $\Sigma\vec{F} = m\vec{a}$

const. acc. Ch2: $v_f^2 = v_i^2 + 2a\underbrace{(x_f - x_i)}_{\Delta x}$

$$\frac{v_f^2 - v_i^2}{2} = a\,\Delta x$$

$$\Sigma W = \Sigma\vec{F}\cdot\vec{s} = m\vec{a}\cdot\vec{s} = m\underbrace{\vec{a}\cdot\Delta\vec{x}}_{\frac{v_f^2 - v_i^2}{2}}$$

$$\Sigma W = \overset{?}{m\frac{v_f^2}{2}} - m\frac{v_i^2}{2} \qquad \left\{ KE \equiv m\frac{v^2}{2} \right\}$$

$$\Sigma W = KE_f - KE_i$$

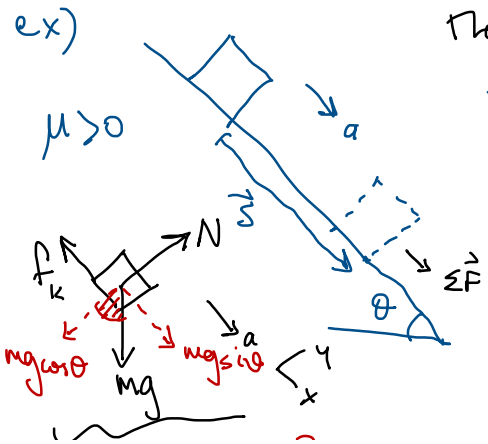
$$\boxed{\Sigma W = \Delta KE}$$

work - kinetic energy Theorem

{Total work done by the net force is equal to the change in KE}

ex)

$\mu > 0$



Find ΣW ; ΔKE? $\boxed{\Delta = f - i}$

$W_f + W_{mg} + W_N$ $\Sigma\vec{F}\cdot\vec{s}$

$\Sigma\vec{F}$

$\Sigma F_y = ma_y^{\,0}$

$N = mg\cos\theta$

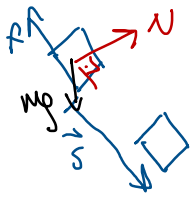
$\Sigma F_x = ma_x$

$mg\sin\theta - f = ma \Rightarrow \underbrace{mg(\sin\theta - \mu\cos\theta)}_{W_{mg} \quad W_f} = ma$

$f = \mu N = \mu mg\cos\theta$

$\Sigma W = \Sigma \vec{F}\cdot\vec{s} = mg(\sin\theta - \mu\cos\theta)\, s\cos 0 = mgs(\sin\theta - \mu\cos\theta)$

$\Delta K = mgs(\sin\theta - \mu\cos\theta) = \frac{mv_f^2}{2} - \frac{mv_i^2}{2}$

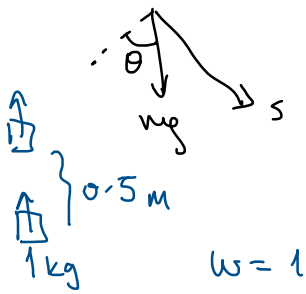


$W_f = \vec{f}\cdot\vec{s} = -fs = -\mu mgs\cos\theta = W_f$

$W_N = \vec{N}\cdot\vec{s} = Ns\cos 90^\circ = 0$

$W_{mg} = m\vec{g}\cdot\vec{s} = mgs\cos(90-\theta)$

$= mgs\sin\theta$

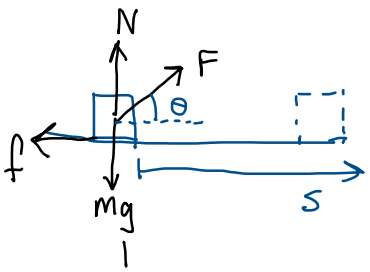


$W = 1kg(10)(0.5) = 5J \simeq 1cal$

$4.184J = 1cal$

---

ex)



$\theta = 37^\circ$

$S = 20m$

$F = 5000N$

$mg = 14700N$

$f = 3500N$

$v_i = 2m/s$

$v_f = ?$

$\Sigma W = \Delta K$

$W_f + W_F + W_N + W_{mg} = K_f - K_i$

$W_F \to 5000(20)\cos 37$, $W_N \to 0$, $W_{mg} \to 0$, $W_f \to 3500(20)\cos 180$

$mg = 14700$

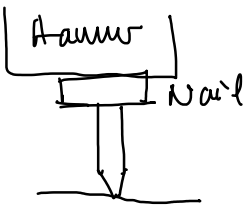
$m = \frac{14700}{9.8} = 1500kg$

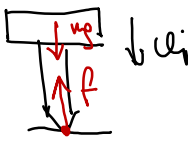
$10000 = \frac{1500}{2}(v_f^2 - 2^2)$

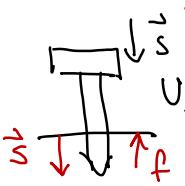
$v_f = 4.2\ m/s$

Hammer and Nail

W done on nail = ΔK







$\Sigma W = W_f + W_{mg} = \Delta K$

$W_{mg} \to 0$

$v_f = 0$

$\Sigma W = \frac{mv_f^2}{2} - \frac{mv_i^2}{2}$ ($v_f^2 \to 0$)

$f \gg mg$

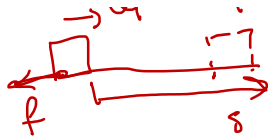
$fs\cos 180 = \Delta K$

$-fs = -\frac{mv_i^2}{2} \Rightarrow \boxed{fs = \frac{mv_i^2}{2}}$

$v_i = 10\ m/s$

$m = 0.01kg$

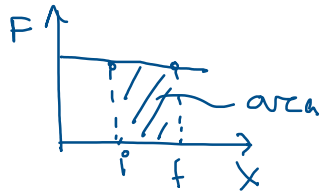
$v_f = 0$

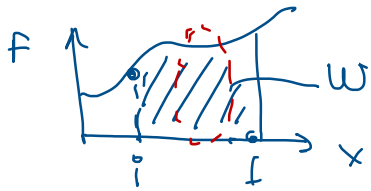


save question

$s = 1\,cm$
$f = ?$

$W = \vec{F}\cdot\vec{s}$ when F const $\Rightarrow$ $a = \text{const}$





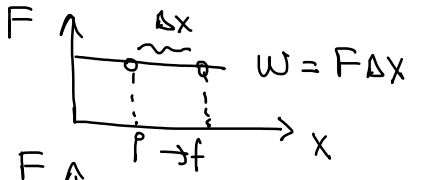
$W = \int \vec{F}\cdot d\vec{x}$

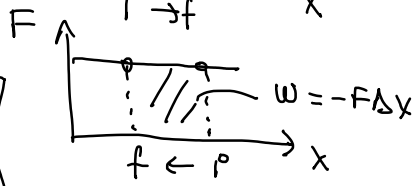
small change

$$W = \int_i^f \vec{F}\cdot d\vec{x} = \int_i^f \vec{F}\cdot d\vec{s}$$

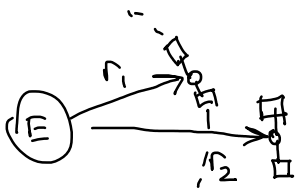
$$W = \int_i^f \vec{F}\cdot d\vec{x}$$

F const $\Rightarrow$ $W = \vec{F}\cdot\int_i^f d\vec{x} = \vec{F}\cdot\Delta\vec{x}$





ex) satellite motion



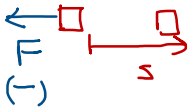
satellite goes from $r_1 \to r_2$;

$W = ?$

$$F = \frac{-1.2\times10^{22}}{r^2}\,\hat{r}$$

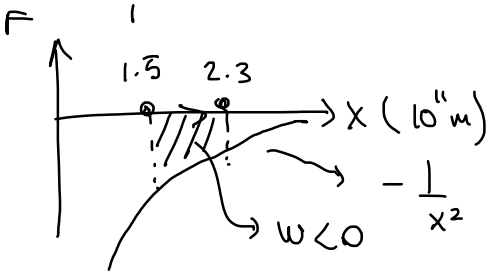
$r_1 = 1.5\times10^{11}\,m$

$r_2 = 2.3\times10^{11}\,m$



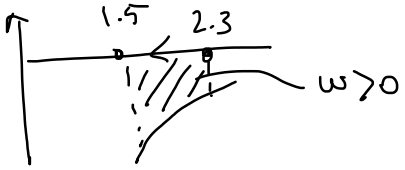
$$F = -\frac{1.2\times10^{22}}{x^2}\;;$$

$$W = \int_i^f \vec{F}\cdot d\vec{x} = 1.2\times10^{22}\int_i^f \frac{dx}{x^2} = -1.2\times10^{22}\left[-\frac{1}{x}\right]_{r_1}^{r_2}$$

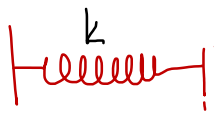


$$= -\frac{1.2\times10^{22}}{10^{11}}\underbrace{\left[-\frac{1}{2.3}+\frac{1}{1.5}\right]}_{(+)}$$

$$W = -3\times10^{11}\,J$$



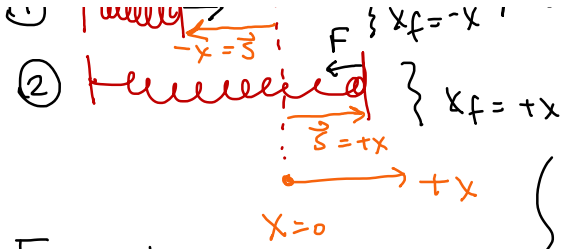
WORK done by SPRING FORCE (yay kuvvetin yaptığı iş)



→ equilibrium point

(1) F

$\vec{F}$ & $\vec{s}$ are opposite



$\vec{s} = \Delta\vec{x} = \vec{x}_f - \vec{x}_i$

$\Downarrow$

x_0 = equilibrium point

$\Downarrow$

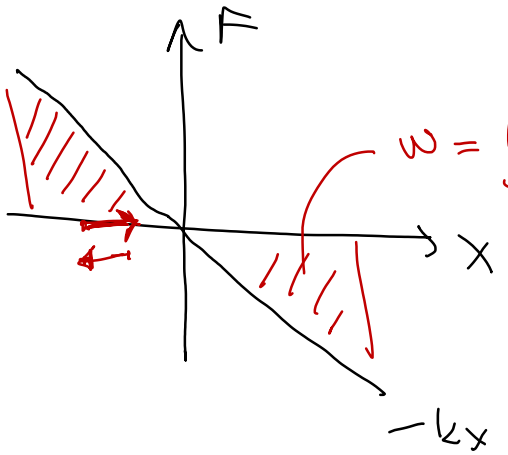
0

①	②
F (+)	F (−)
−x	+x

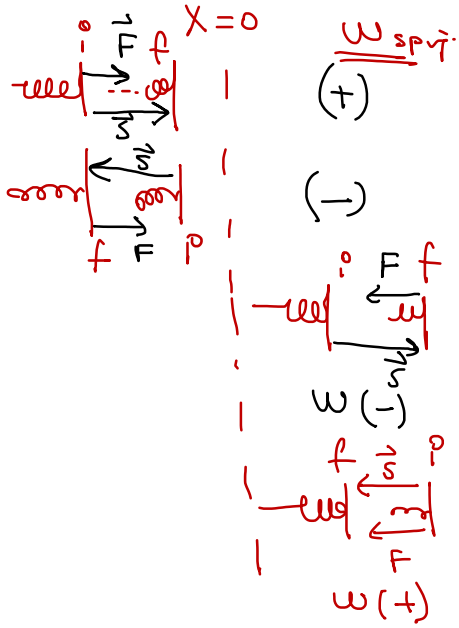
$F = -kx$

$x \Rightarrow +x \qquad F = -kx$

$x \Rightarrow -x \qquad F = -k(-x) = +kx$



$$W = \int_i^f \vec{F}\cdot d\vec{x}$$



work by spring

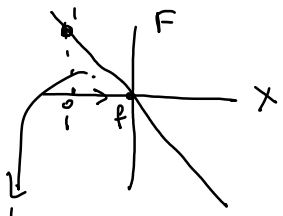
$$W = \int_i^f \vec{F}\cdot d\vec{x} = \int_i^f -kx\,dx = -$$

$$W_s = -k\left[\frac{x_f^2}{2} - \frac{x_i^2}{2}\right] = k\frac{x_i^2}{2} - \frac{kx_f^2}{2}$$

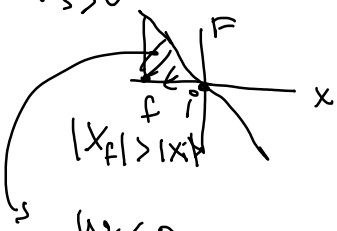
$W_s < 0 \qquad |x_f| > |x_i|$

$W_s > 0 \qquad |x_i| > |x_f|$

$$W_s = k\frac{x_i^2}{2} -$$



$W_s > 0$



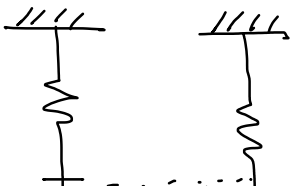
$W_s < 0$

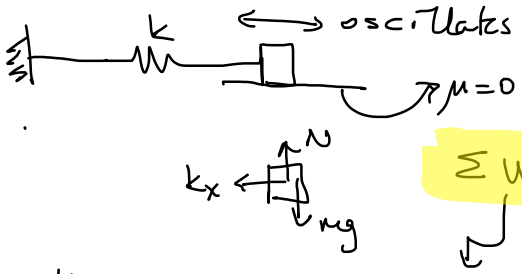
$F = -kx \qquad k = \left[\frac{N}{m}\right] \qquad W \sim \frac{kx^2}{2} \qquad k =$

ex) $k = 500\,N/m \qquad x_i = 0.2\,m \qquad x_f = 0.5\,m$

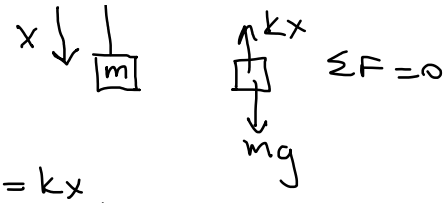
$$W_s = k\left(\frac{0.2^2}{2} - \frac{0.5^2}{2}\right) = -52.5\ J$$

How to measure k?





Σ



$\Sigma F = 0$

$mg = kx$

$$\boxed{k = \frac{mg}{x}}$$

$$W_s + \cancel{W_g}^{\,0} + \cancel{W_N}^{\,0}$$

$$\left(k\frac{x_i^2}{2} - k\frac{x_f^2}{2}\right)$$

related to

Power (Giq)

$$P \equiv \frac{\text{Energy}}{\text{time}} \Rightarrow \left[\text{Watt} = \frac{\text{Joules}}{\text{sec}}\right] \qquad P = \frac{W}{t}$$

Engines $\Rightarrow$ 1 hp = 1 horsepower = 746 W

100 hp = 746 ≅ 7

Electric bills: 1 kWsa = 1 kW hr

10^3 W 3600 s = 3.6×10^6 J

$$\boxed{P_{ave} = \frac{\Delta W}{\Delta t}} \; ; \quad P = \frac{dW}{dt} = \frac{\vec{F}\cdot d\vec{x}}{dt} = \vec{F}\cdot\frac{d\vec{x}}{dt}$$

$W = \int F\,dx$

$dW = F\,dx$

$$\boxed{P = \vec{F}\cdot\vec{v}}$$

ex) Airbus plane

$F = 322\,000$ N

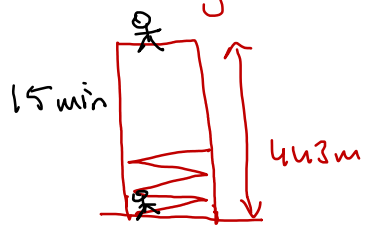
$v = 250$ m/s

$P = 322 \times 250 \times 10^3$

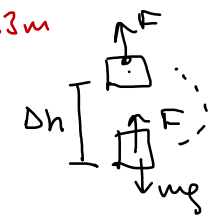
$P = 8.05 \times 10^7$ W

$P = 108\,000$ hp

ex) a 50 kg runner, runs up 443 m tall tower in 15 minutes



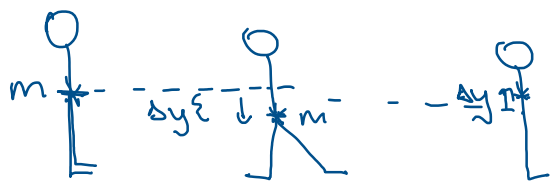
$$P = \frac{W}{t} = \frac{\overset{mg}{(F)}\,\Delta h}{15 \times 60\,s} = \frac{mgh\ (\equiv PE)}{t} =$$



$P = 241$ W $= 0.241$ kW

$= 0.323$ hp

work when we walk?



center of mass

$\uparrow F\,\Delta y$ = each step work

mg (5cm)

mg 0.

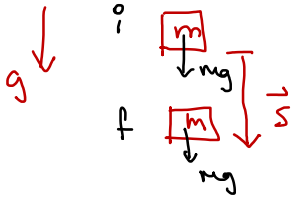
50 (g

calories $\leftrightarrow$ joules

→ End of Ch 6 →

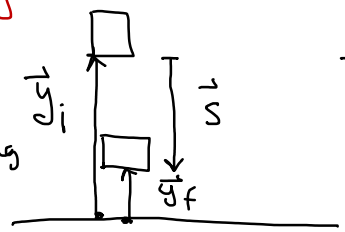
CHAPTER 7

Potential Energy & Energy Conservation



$W_{mg} = m\vec{g}\cdot\vec{s}$

$W_{mg} = m\vec{g}\cdot(\vec{y}_F - \vec{y}_i)$

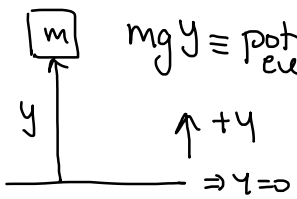


$\vec{s} = -(\vec{y}_i$

$= m\vec{g}\cdot\vec{y}_F - m\vec{g}\cdot\vec{y}_i$

$= mg\,y_F \underbrace{\cos 180}_{(-1)} - mg\,y_i \underbrace{\cos 180}_{(-1)}$

$W_{mg} = \underbrace{mg\,y_i}_{\text{Something = potential energy}} - mg\,y_F = U_i - U_F = W_m$



$mgy \equiv$ pot. energy

$PE = U = mgy$

Δ = final − initial

$U_i - U_F = W_{mg}$

$\boxed{-\Delta U = W_{mg}}$ ***

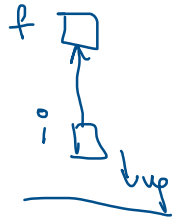
Ch6

$\Sigma W = \Delta$

$W_{mg} > 0$

$-\Delta U > 0$

$U_i > U_F$



$W_{mg} < 0$

$-\Delta U < 0$

$U_i < U_F$

Ch6

$\Sigma W = \Delta K$

If $\Sigma W = W_{mg}$

?! only mg is doing work !!

Ch7

$W_{mg} = -\Delta U$

$$\Sigma W = W_{mg}$$

$$\Delta K = -\Delta U$$

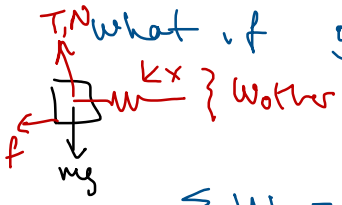
$$K_F - K_i = -(U_F - U_i) = U_i - U_F$$

$U =$

$K =$

$$* \; K_F + U_F = U_i + K_i$$

} Energy conserv when only doing wo



What if other forces are also doing work

↳ friction, Tension, Normal, spring force.

$$\Sigma W = W_{mg} + W_{others} = \Delta K$$

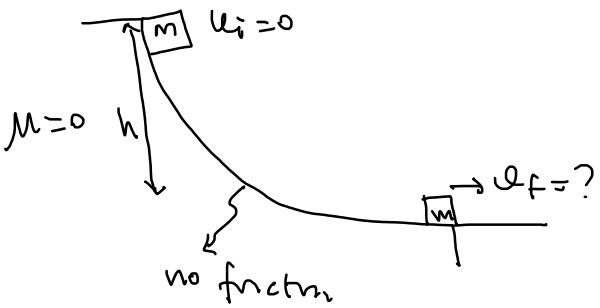
$U = U_g = mg$

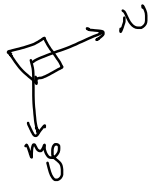
$$-\Delta U_g + W_{others} = \Delta K$$

$$U_i - U_F + W_{other} = K_F - K_i$$

$$W_{other} + U_i + K_i = U_F + K_F$$

} Energy co for all sh

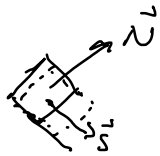




$$W_{other} + U_i + K_i = U$$

$$\underline{\underline{W_N}} \to 0$$

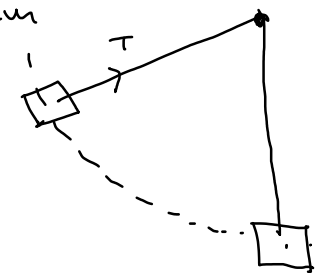
$$W_N = ?$$

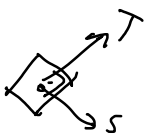


$$\vec{N} \cdot \vec{s} = W_N$$

$$N s \underbrace{\cos 90^\circ}_{0} = 0$$

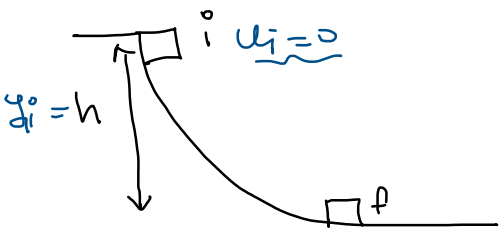
pendulum





$$W_T = \vec{T} \cdot \vec{s} = T s \cos 90 = 0$$

$$W_{other} = W_T = 0$$



$$W_{other} + U_i + K_i = U_f + K_F$$

$$\downarrow$$

$$0 \quad mgy_i + \frac{mv_i^2}{2} = mgy_F + \frac{mv_F^2}{2}$$

$\to y = 0 = y_F$

ex

$$mgh + 0 = mg(0) + \frac{mv_F^2}{2}$$

$$v_F = \sqrt{2gh}$$

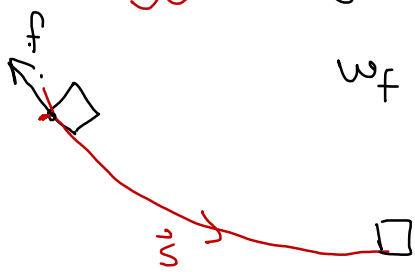
$h = 10m$

$$v_F = \sqrt{2(9.8)10} = 14$$

$W_f = ?$ $\qquad m = 1kg$

$$W_{other} + U_i + K_i = U_F + K_F$$

$$W_f + mgy_i + \frac{m\cancel{v_i^2}^{\,0}}{2} = mg\cancel{y_f}^{\,0} + \frac{mv_F^2}{2}$$



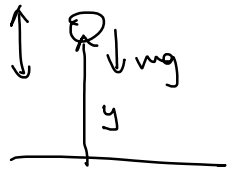
$$W_f = \frac{mv_F^2}{2} - mgy_i = 1\,\frac{5^2}{2} - 1(9.8)10 =$$

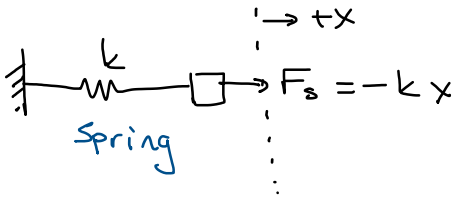
$$W_f = \underbrace{\int f\,dx}_{\text{Hard}}$$

$$= \bar{f} \cdot s$$

f

$$W_{mg} = -\Delta U_g \qquad U = mgy$$





$$W_s = \text{elastic work} = \frac{1}{2}kx_i^2 - \frac{1}{2}kx_f^2$$

$$\int \vec{F}\cdot d\vec{x} = W = \int_i^f -kx\,dx =$$

$$W_s = \frac{1}{2}kx_i^2 - \frac{1}{2}kx_f^2$$

?! $\quad \frac{1}{2}kx^2 \equiv U_{el}$ = elastic potential

$$W_s = U_i - U_F = -\Delta U_{el}$$

$$\Sigma W = \Delta K \quad \Rightarrow$$

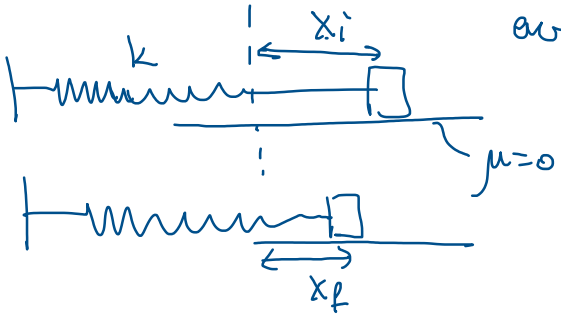
others $\quad$ spring $\quad$ mg

W_{others} $\quad$ $W_s = -\Delta U_{el}$ $\quad$ $W_{mg} = -\Delta U_g$

$$W_{others} + U_{el,i} - U_{el,f} + U_{g,i} - U_{g,f} = K$$

$$\boxed{W_{other} + U_{el,i} + U_{g,i} + K_i = U_{el,f} + U_{g,f} + K_f}$$

ex)



energy conservation

spring is stretched 0.1 m

$v_i = 0$ $\quad k = 5\ N/m \quad m =$

$v_f = ?$ at $x_f = 0.08\ m$

no motion in y $\Leftrightarrow$ $\Delta U_g = 0$

$$\cancel{W_{other}} + U_{el,i} + \cancel{U_{g,i}} + K_i = U_{el,f} + \cancel{U_{g,f}} + K_f$$

$\mu = 0$

$$\frac{1}{2}kx_i^2 + \frac{1}{2}m\cancel{v_i^2}^{\,0} = \frac{1}{2}kx_f^2 + \frac{1}{2}mv_f^2$$

$$\cancel{\tfrac{1}{2}}\,5\,(0.1)^2 = \cancel{\tfrac{1}{2}}\,5\,(0.08)^2 + \cancel{\tfrac{1}{2}}\,0.2$$

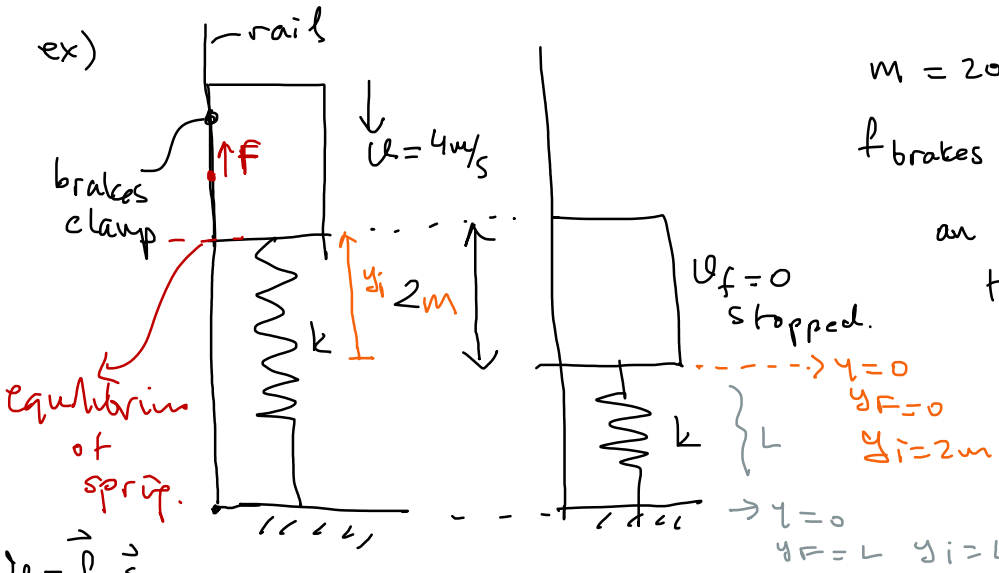
$$0.09 = v_f^2$$

$v_f = +0.3\ m/s$ ✓

$v_f = -0.3\ m/s$ ✓

A 2000-kg (19,600-N) elevator with broken cables in a test rig is falling at 4.00 m/s when it contacts a cushioning spring at the bottom of the shaft. The spring is intended to stop the elevator, compressing 2.00 m as it does so (Fig. 7.17). During the motion a safety clamp applies a constant 17,000-N frictional force to the elevator. What is the necessary force constant k for the spring?

ex)



$m = 2000\ kg$; $mg = 1$

$f_{brakes} = f_{friction} = 170$

an elevator brakes o

to stop ; spring i

compressed 2m

the elevator

$k = ?$

$W_f = \vec{f} \cdot \vec{s}$

$= -fs$

$$W_{other} + U_{el,i} + U_{g,i} + K_i = U_{el,f} + U_{g,f} + K_f$$

$$-fs + \frac{1}{2}k(\Delta y_i^2) + mgy_i + \frac{1}{2}mv_i^2 = \frac{1}{2}k(\Delta y_f)^2 + mgy_f +$$

$$-17000(2) + 0 + 19600(2) + \frac{1}{2}2000\,4^2 = \frac{1}{2}k(2)^2 + 19600(0)$$

$$k = 1.06 \times 10^4\ N/m$$

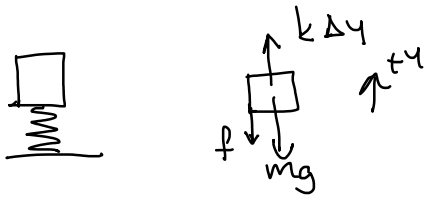
⇒ if there was no frictional force; everything else is same

Case 2

$$39200 + 16000 = \frac{1}{2}k'\,2^2 + 0$$

$$k' = 2.76 \times 10^4\ N/m$$

In case (I) when elevator stops @ 2m; with it move up



If $\Sigma F \leq 0$ it won't move up!

$$\Sigma F = 10600(2) - 19600 - 17$$

$$\underset{21200}{10600(2)} - 19600 - f$$

$$f = 1600\,N$$

$F = -\Delta U$?

FORCES

CONSERVATIVE (korunumlu)

NON CONSERVATIVE (-suz)

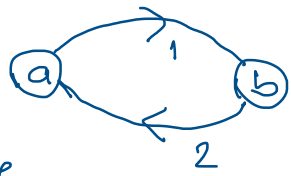
$$\left\{ \begin{array}{l} mg \ ; \ F_s = -kx \ ; \ k\frac{q^2}{r^2} = F \\ G\frac{m_1 m_2}{r^2} = F_g \end{array} \right\}$$

$f = \mu N$ ∴ frictional are non cov

properties of work done by conservative forces.

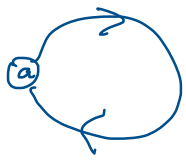
I) $F = -\Delta U$ can written like this.

II) It's Reversible (tersinir)



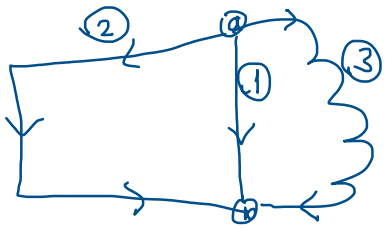
$$W_{a\to b} = W_1 = -W_2 = W_{b\to a}$$

III) if a=b save

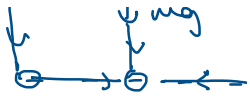


$W = 0$

IV It's independent of the path.



$W_1 = W_2 = W_3$



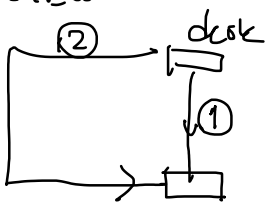
Nonconservative ones: irreversible ; dependent of p

frictional forces

$\cancel{F = -\Delta U}$

$W \neq 0$

TOP VIEW



$W_2 > W_1$

$W_f = \vec{f} \cdot \vec{s} \leq 0$

$\vec{f} \cdot \vec{s} = 0$

↳ static friction

$W_f < 0 \Rightarrow$ friction steals energy from you!

amount energy lost in friction can go to → shape c

→ increase molecule

↳ t

$F = -\Delta U$

$\downarrow$

$\vec{F} \cdot \Delta\vec{x}$

$F \Delta x = -\Delta U$

$F = -\frac{\Delta U}{\Delta x}$

$F = -\frac{dU}{dx}$

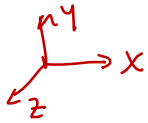
$d \Rightarrow$ differential

$\partial \Rightarrow$ partial differential. (kismi)

$F_x = -\frac{\partial U}{\partial x}$ $F_y = -\frac{\partial U}{\partial y}$ $F_z =$

$$\boxed{\vec{F} = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}}$$

$U = mgy$



$\frac{\partial U}{\partial x} = 0$

$\frac{\partial U}{\partial z} = 0$

$\frac{\partial U}{\partial y} = mg$ $-\frac{\partial U}{\partial y} = F_y = -mg$ $\downarrow$ $mg = F_y$

$\vec{F} = -$

$\vec{F} = -kx\,\hat{i}$

ex $U = 5x^2 - 3x$

$U =$

$U = -\frac{1}{2}kx^2$ $\quad -\frac{\partial U}{\partial x} = F_x = -\frac{1}{2}k\,2x = -kx$

$\frac{\partial U}{\partial z} = \frac{\partial U}{\partial y} = 0$

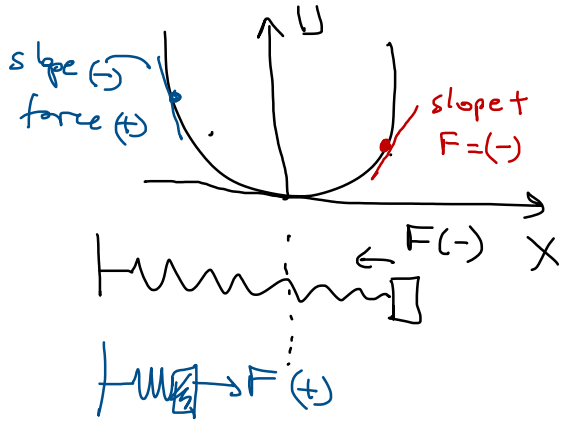
$\vec{F} = ?$

$-\frac{\partial U}{\partial x} = -(10x - 3$

$-\frac{\partial U}{\partial y} = -(8y$

$-\frac{\partial U}{\partial z} = -($

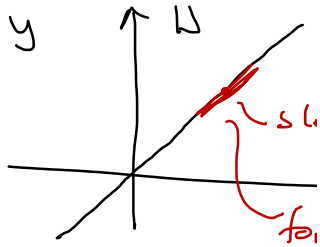
F = derivative of U w.r.t. x (y or z)

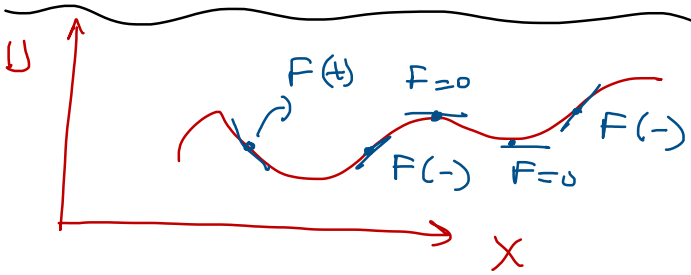


$F = -\frac{\partial U}{\partial x} = -(\text{slope})$

derivative slope

$U = mgy$



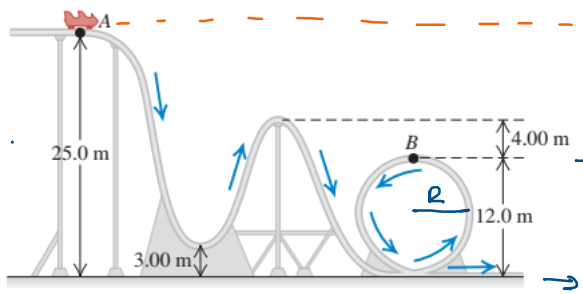


End of Ch7

7.45 •• A 350-kg roller coaster starts from rest at point A and slides down the frictionless loop-the-loop shown in Fig. P7.45. (a) How fast is this roller coaster moving at point B? (b) How hard does it press against the track at point B?

Figure **P7.45**

$m = 350\ kg$



$y_A = 13m$

$-12 - y = 0$

$y_B = 0$

$\rightarrow y = 0$

$v_y = 0$

$w.f = 0$

$E_A = E_B$

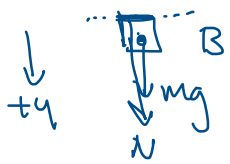
$\cancel{mg}\,y_A + \cancel{m}\frac{v_A^2}{2} = \cancel{mg}\,y_B + \cancel{m}$

$(9.8)(13) + 0 = 0 + \frac{v_B^2}{2}$

$v_B = \sqrt{2(9.8)13} =$

b) $N = ?$

$a = \frac{v_B^2}{R} = \frac{15.96^2}{6} =$

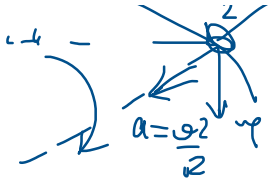
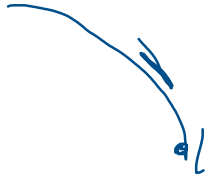


$\Sigma F = ma$

$N + mg = ma$

$N + mg = m\frac{v^2}{R} \Rightarrow N = m\left(\frac{v^2}{R} - g\right)$

$N = 11433.3$ Newton


$a = \frac{v^2}{R}$